

Woojun Jung

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RESEARCH INTERESTS

Representation Learning: learning representations from heterogeneous data and interaction histories for adaptive memory and long-horizon personalization.

Human-AI Interaction: personalized AI systems that support human understanding and decision-making.

EDUCATION

Korea University

Mar 2025 – Feb 2027 (*Expected*)

M.S. Student in Computer Science and Engineering

Advisor: Prof. [Susik Yoon](#)

Korea University

Mar 2022 – Feb 2025

B.E. in Computer Science and Engineering

Korea University

Mar 2016 – Feb 2022

B.F.A. in Industrial and Information Design

2 years of Military Service: Republic of Korea Air Force (2017 – 2019)

Relevant Coursework (all listed courses A/A+): *AI/ML* - Deep Learning, Optimization Theory, Natural Language Processing, Reinforcement Learning, Artificial Intelligence, Brain and Machine Cognition; *CS Foundations*-Algorithms, Computer Architecture; *Design/HCI* - Interaction Design, Interface Design, Design and Human Factors, Design Methodology.

PUBLICATIONS

**: equal contribution, †: corresponding author*

Peer-reviewed Papers

- [3] **Segment-driven Structural Induction and Semantic Alignment for Heterogeneous Tabular Representation**

[Woojun Jung](#) and [Susik Yoon](#)[†]

ICML 2026 — [\[arXiv\]](#) [\[Code\]](#)

- [2] **Can Structural Cues Save LLMs? Evaluating Language Models in Massive Document Streams**

Yukyung Lee, Yebin Lim*, [Woojun Jung](#)^{*}, Wonjun Choi, [Susik Yoon](#)[†]

KDD 2026 — [\[ACM DL\]](#) [\[Code\]](#)

- [1] **HAETAE: In-Domain Table Pretraining with Header Anchoring**

[Woojun Jung](#) and [Susik Yoon](#)[†]

SIGIR 2025 — [\[ACM DL\]](#) [\[Code\]](#)

RESEARCH EXPERIENCE

DAIS Lab, Korea University

Aug 2024 – Present

Research Intern (Aug 2024 – Feb 2025); Research Assistant (Mar 2025 – Present)

Calibrated User Beliefs for Conversational Agents

May 2026 – Present

- Formulating long-horizon personalization as maintaining calibrated beliefs over context-dependent and evolving user preferences rather than reducing them to fixed textual point memories.
- Exploring whether representation-derived uncertainty can be calibrated against subsequent user feedback and used to guide when agents should apply a preference or ask for clarification.

Structured Understanding of Tabular Data

ICML 2026; SIGIR 2025

- Led two first-author projects on domain-specific tabular representation learning.
- Proposed header anchoring to regularize contextualized header embeddings toward context-independent anchors, stabilizing semantics across row contexts and schema variants.

- Developed a pretraining method that jointly learns schema and column contexts with entropy-routed objectives to preserve domain-coherent and instance-specific attribute semantics.

Structured Understanding in Evolving Information Streams

KDD 2026

- Developed the Temporal QA task, one of three benchmark tasks alongside Topic Clustering and Summarization.
- Designed and implemented its evaluation protocol, benchmark dataset, and human-annotation guidelines.

APPLIED RESEARCH COLLABORATION

Foundation Models for Enterprise Relational Data, KT (Korea Telecom)

Mar 2025–Present

- Collaborated with KT on foundation-model methods for enterprise relational databases.
- Analyzed challenges in adapting tabular ML to organizational data with heterogeneous schemas.
- Prototyped a pretraining pipeline for schema-aware attribute representations for enterprise tables.
- Investigating schema-linking methods that map natural-language questions to relevant tables and columns.

SELECTED PROJECTS

Projects translating AI into interactions that support human understanding, agency, and decision-making.

Dr.Snap: Multimodal Dermatology Decision-Support System

2024

- Designed a questionnaire-driven, multi-turn interaction that helps users communicate symptoms and receive structured decision support.
- Developed an end-to-end user flow from symptom intake and examination to assessment and follow-up guidance.
- Implemented a multimodal prototype combining image classification, retrieval-augmented generation, and conversational interaction; received Third Place at the Korea University Capstone Design Competition.

Neuro: BCI-Enabled VR Interaction Concept

2021

- Explored how neural signals could be translated into meaningful feedback, creative expression, and user control in virtual environments.
- Designed the wearable form factor, interaction scenarios, and visual prototypes for an approachable everyday BCI experience.

TEACHING EXPERIENCE

Teaching Assistant, Data Science, Korea University

Spring 2025

- Prepared and delivered tutorials on LaTeX and visualization with matplotlib for undergraduate students.

Teaching Assistant, Artificial Intelligence, Korea University

Fall 2025

- Prepared and delivered a tutorial on multimodal representation learning with CLIP for undergraduate students.

AWARDS & SCHOLARSHIPS

Master's Fellowship (KRW 12,000,000), National Research Foundation of Korea (NRF)

2025–2026

Third Place, Korea University Capstone Design Competition

2024

Third Place, Samsung Design Competition

2019

SKILLS

Programming: Python, C, SQL, Git, GitHub, LaTeX, HTML, CSS, JavaScript

ML & Systems: PyTorch, CUDA, Transformers, vLLM, NumPy, pandas, scikit-learn, matplotlib

Design & Hardware Prototyping: Figma, Illustrator, Photoshop, Rhinoceros, KeyShot, 3D printing, Arduino

Languages: Korean (Native), English (Fluent)